Online Supplementary Appendices

"The Use of Quantile Methods in Economic History"

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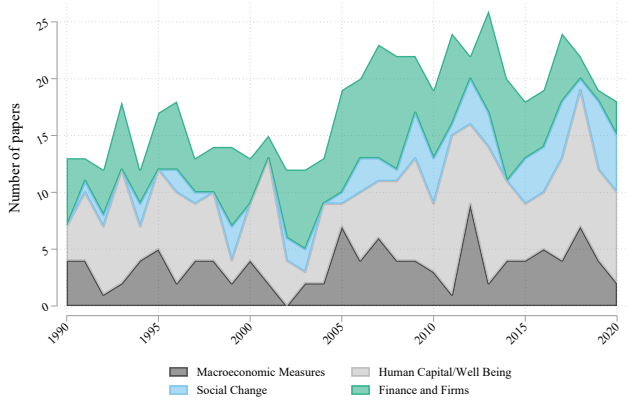
A A Descriptive Analysis of Studies in a Single Economic History Journal

Reviewing cases where quantile regression *is* used, as we did in the body of this paper, suggests the broad applicability of these methods across themes and settings in economic history, but does not indicate the additional scope for use in papers which focus principally on mean effects. To see the *potential* for Quantile Regression, we carried out a review based on all papers published over the last 30 years in a specific economic history journal (Explorations in Economic History, hereafter EEH). This allows us to gain a more complete picture of trends in published research in a specific highly-cited economic history journal, consider the methods used, and the potential for distributional analyses in economic history. In this review, we read each paper published over the last 3 decades (the first volume of 1990 to the last volume of 2020), and classified each paper according to whether it contained empirical methods, or was theoretical. In the case of empirical papers, we then classified these by method (OLS, Logit, Quantile Regression, and so forth) and finally whether principal dependent and independent variables are continuous, discrete, ordinal, etc. We note that while such an analysis is useful in understanding broad patterns and *pontential* applicability of quantile methods, the true use of such methods requires a case-by-case analysis of exact data availability, sample sizes, modelling considerations, and so forth.

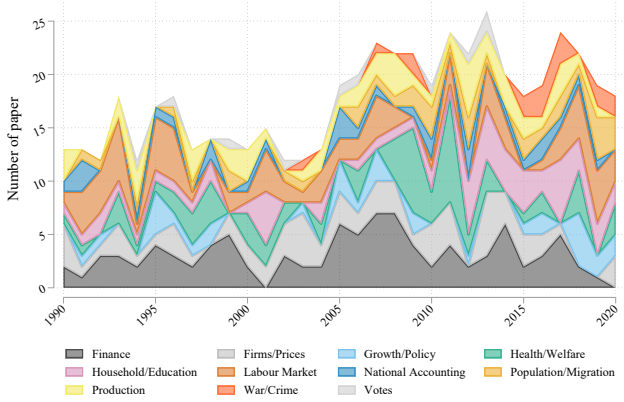
Prior to considering trends in measures and methods, it is useful to provide descriptive evidence of the key themes studied over the past 30 years in this particular journal. In Figure A.1, we plot the broad areas which papers study, as classified based on our reading of the literature. These classifications refer to the dependent variable under study in all empirical papers published in EEH. The right-hand plot classifies papers into one of four aggregate areas: macroeconomic measures, human capital and well-being, social change (which includes papers studying topics such as conflict, migration, and changing political landscapes), and finance and firms. In general we note the largest of these groups is that related to microeconomic areas such as education, health and labour markets (the shaded light gray area), followed by themes related to finance and firms, with relatively less work focused principally on macroeconomic

Figure A.1: Trends in the dependent variable in papers published in *Explorations in Economic History*

(a) Large groups



(b) Small groups



Notes: Based on our reading of each paper published in EEH we classify each paper as belonging to a particular (single) area, as described in the legend in panel (b). These classes are further aggregated in panel (a), where 'Macroeconomic Measures' refers to Growth/Policy, National Accounting and Production; 'Human Capital/Well Being' refers to Health/Welfare, Household/Education and Labour Market; 'Social Change' refers to Population/Migration, War/Crime and Votes; and Finance and Firms refers to Finance and Firms/Prices. A file summarising all papers in terms of a number of characteristics discussed in this review is provided as an online appenix to this paper.

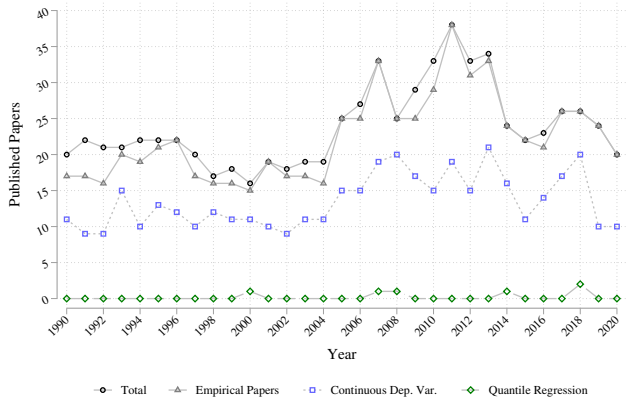
indicators. In the left-hand panel we provide more disaggregated classes, seeing more noisy patterns, and evidence of broad coverage of topics within economic history.

While it is illustrative to observe trends in the published record in economic history, here we are most interested in considering how the types of questions and measures used in these papers relate to their potential for using quantile methods. At minimum, this implies that dependent variables need to be continuous, given that quantile methods examine impacts of some independent variable across the distribution of the dependent variable.

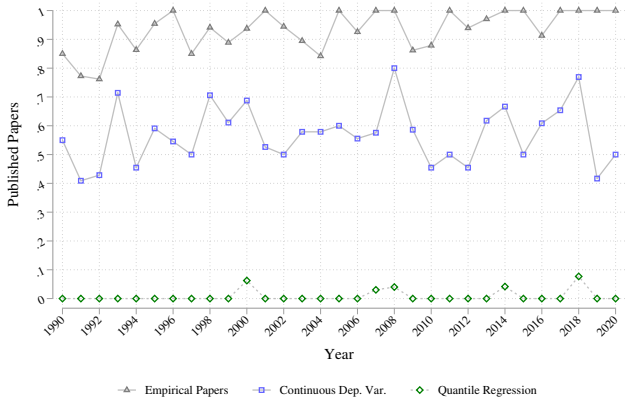
Figure A.2 plots the number and share of papers published in *EEH*, classifying them by whether they are empirical or not, and in the case of empirical papers, whether the dependent variable is continuous, and hence suitable for quantile analyses. Finally, we plot the number and share of papers that effectively do undertake quantile analyses. In Figure A.2 panel (a) we can see how the amount of total articles published in *EEH* was around 20 annual publications, with an increase to around 30 from 2005 onwards. We can see how the number of empirical papers follows a similar trend to that of total articles, and corresponds to 93.1% on average of the number of total articles. In recent years, this value is even higher, with 100%, or close to 100% of articles containing at least some considerable empirical portion, pointing to the growing frequency of empirical evidence in studies in economic history.

Figure A.2: Trends in Publications in an Economic History Journal

(a) All Papers



(b) Proportion of Papers



Notes: The total number of papers (panel A) and the proportion of all papers (panel B) refer to all papers in the journal EEH, based on the isssue in which they are published. This data was collected based on a reading of each paper and classification of the methods and 'principal' dependent variables.

While this is illustrative of the importance of quantitative analyses in understanding historical processes in economics, it does not necessarily imply a suitability for analysis with quantile methods. However, if we observe the articles that are based on continuous dependent variables and hence amenable to quantile-based methods, we observe that they correspond to about 56.5% of the quantity of total articles (over the past 30 years). Despite these two stylised facts: (i) that there is considerable quantitative evidence brought to bear in economic history papers, and (ii) that many of these analyses are based on variables which have continuous distributions and hence potentially interesting distributional features, we observe little use of quantile methods. Among the 738 papers examined, 692 of which were empirical, only 6 studies used quantile regression (as described in the previous sub-section). Unless there is a particular interest in the mean, these facts taken together point to a considerable margin for the adoption of other estimation methods such as quantile regression or QTE methods within economic history.

Finally, we note how these continuous dependent variables are situated within analyses. In Table A.1 we provide a full break down of all the classes of dependent variables observed across all studies from EEH. 418 papers had a single continuous outcome variable of interest, while another 29 had a continuous outcome variable as well as other non-continuous measures. Of these 447 studies, 201 had independent variables which were also continuous, and so more suited to quantile regression methods, while 33 had a binary independent variable of interest, and so are potentially well suited to QTE methods. Many other studies have various types of independent variables, and as such either QTE or quantile regression methods may be appropriate.

Table A.1: Classification of Dependent and Independent variables

	Dependent		Independent	
Type	Number	Proportion	Number	Proportion
Continuous	418	76.70	207	37.91
Binary	58	10.64	47	8.61
Discrete	23	4.22	12	2.20
Ordinal	3	0.55	0	0
Multiple	10	1.83	115	21.06
Continuous-Binary	19	3.49	77	14.10
Continuous-Discrete	9	1.65	21	3.85
Continuous-Dummy	0	0	25	4.58
Continuous-Ordinal	1	0.18	0	0
Dummy-Dummies	0	0	30	5.49
Discrete-Binary	4	0.73	12	2.20

Notes: Multiple classification is article with more than two types of variables. The classification that presents 2 types of variables, correspond to papers that present more than one regression in which its dependent variables are continuous plus one or more variables of another type, and both regressions are relevant in the research.

B Further Empirical Details Related to Demographics and Economic Growth

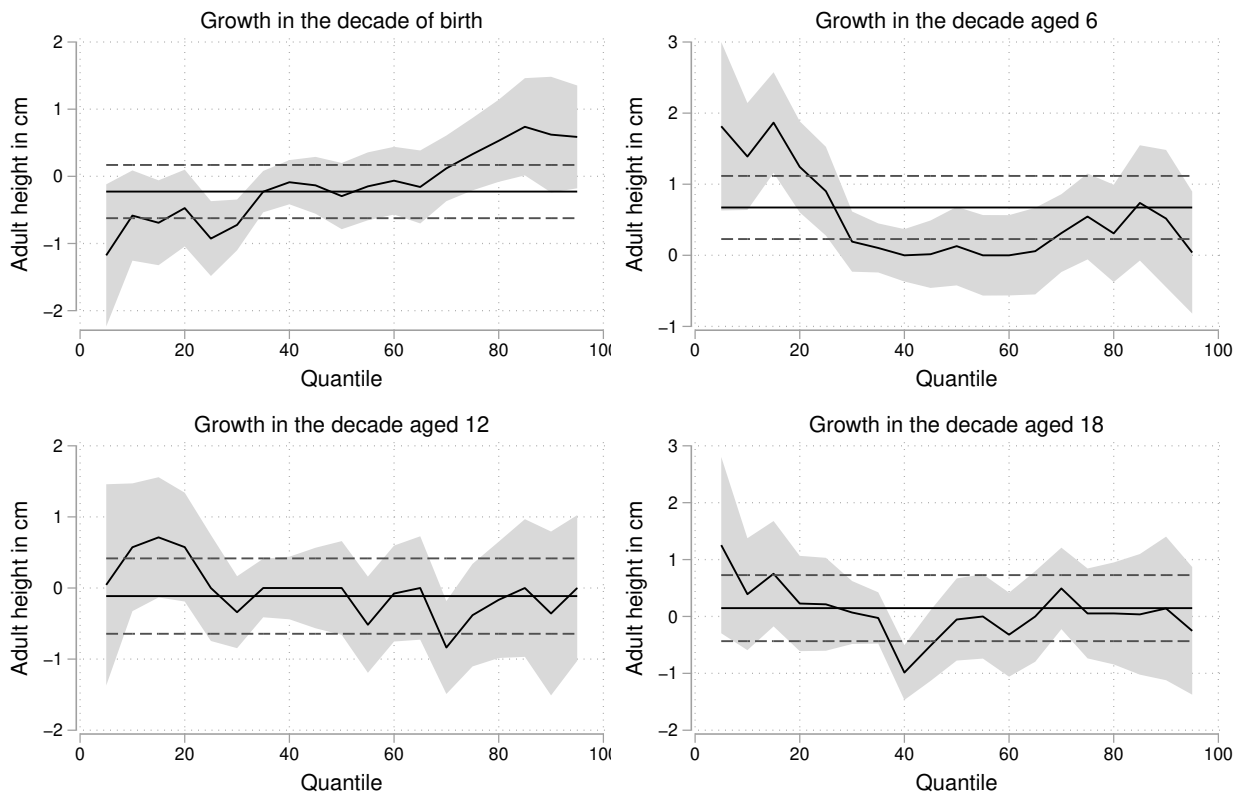
In the body of the paper, we proposed an analysis whereby individuals were associated with prevailing rates of economic growth in the decade in which they were aged 0, 10 and 20. Here we consider a model in which more fine-grained exposure periods are used. Namely, instead of estimating equation 5 based on 10 year age groups, we estimate:

$$\text{Height}_{ipt} = \beta_0 + \beta_1 \text{Growth}^0_{pt} + \beta_2 \text{Growth}^6_{pt} + \beta_3 \text{Growth}^{12}_{pt} + \beta_4 \text{Growth}^{18}_{pt} + \mu_p + \lambda_t + \varepsilon_{pt}. \quad \text{(B.1)}$$

Here all details follow equation 5 in the main text, however Growth^6 refers to the rate of growth when the individual was aged 6, Growth^{12} refers to the rate of growth when individuals were aged 12, and Growth^{18} refers to rates of growth when individuals were aged 18. While this allows us a more fine-grained measure of exposure, it is not without complications. Specifically, in this case note that for individuals born in early years of a decade, rates of growth when aged 0 will be identical to rates of growth when aged 6. Similarly, for individuals born later in a decade, rates of growth when aged 6 will be identical to rates of growth when aged 12, as both will occur in the decade following their birth. In general, this smoothing of rates of growth seems likely to reduce the signal in independent variables, despite the fact that it does introduce variation in ages not identically captured in equation 5 of the main

text.

Figure B.1: Quantile Regression



Notes: Each sub-plot presents the OLS and quantile regression estimates of the effect of local rates of economic growth at particular points of an individual's life on their adult height. Solid black lines and dashed lines refer to OLS point estimates and 95% CIs, while gray solid lines and shaded areas refer quantile regression point estimates and 95% CIs.

Results, both of quantile and OLS models following the structure discussed in section 5 of the paper are provided in Figure B.1. The top-left panel documents a null effect of exposure to growth in the decade of birth on adult height when considering average effects, although marginally negative effects lower in the distribution of adult height, and marginally positive effects higher in the distribution. Clear heterogeneity in the impact of growth is observed in the top right-hand panel. While exposure to growth in the decade in which a child is aged 6 years is observed to significantly increase adult height (OLS estimates suggest a 1% higher rate of growth is associated with nearly 1cm in additional height), this effect is very heterogeneous, with large impacts at the lower end of the distribution of height, and null impacts among taller individuals. This is similar to the pattern described in the body of the paper, with similar implications. While some similar patterns may even persist when children are older (bottom panels), these effects are observed to be most notable in the early years of life when children are particularly sensitive to their conditions. Results in the bottom two panels suggest quite flat results across the

distribution of adult height when considering growth at older ages. From this we can infer that results from standard linear models estimated by OLS are not hiding considerable heterogeneity in the effect of these particular variables.